

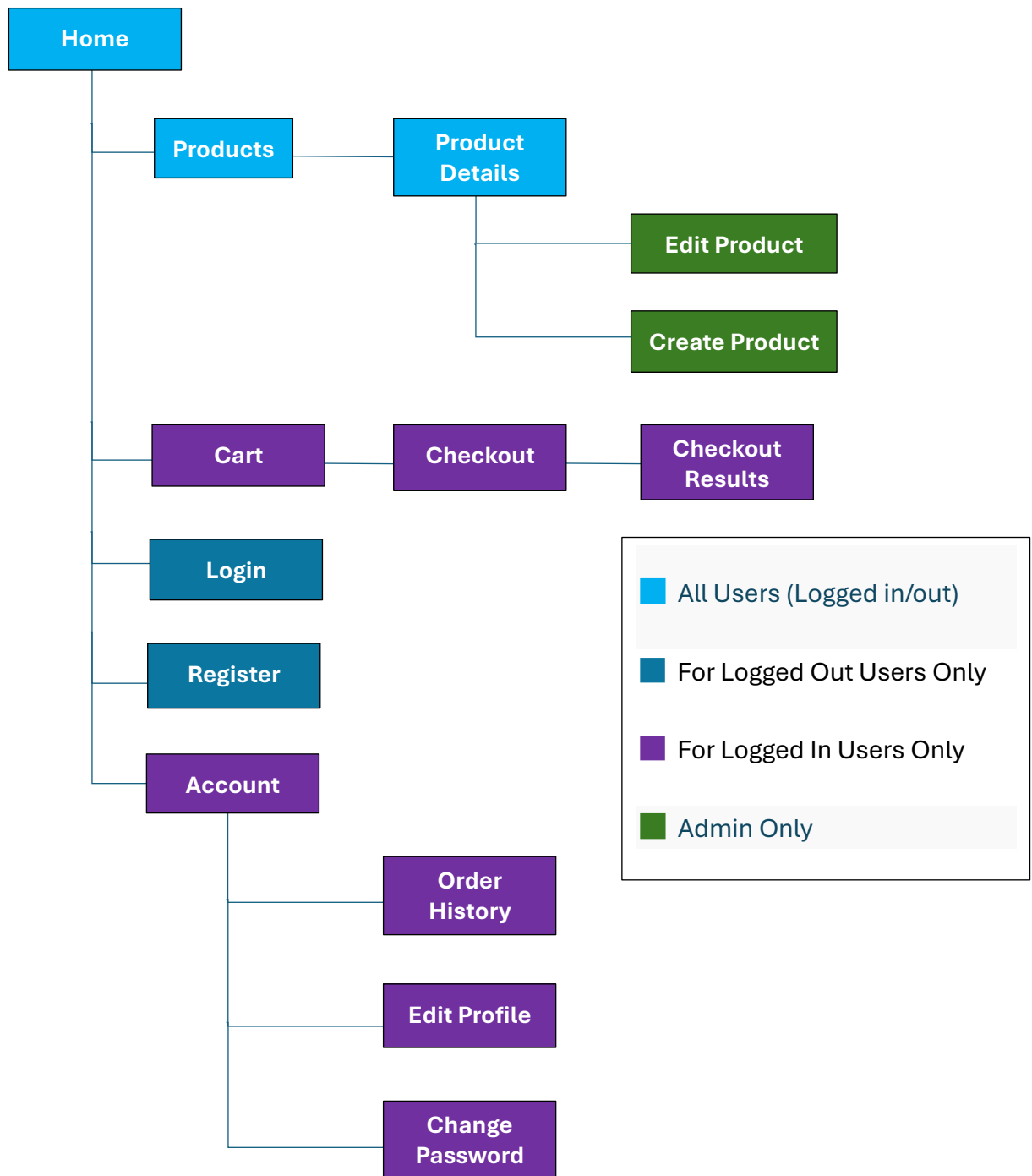
SE 352/452 - Object-Oriented Enterprise Computing

Final Project: Full-Stack Spring Boot Application: “Gearwood Puzzle Store”

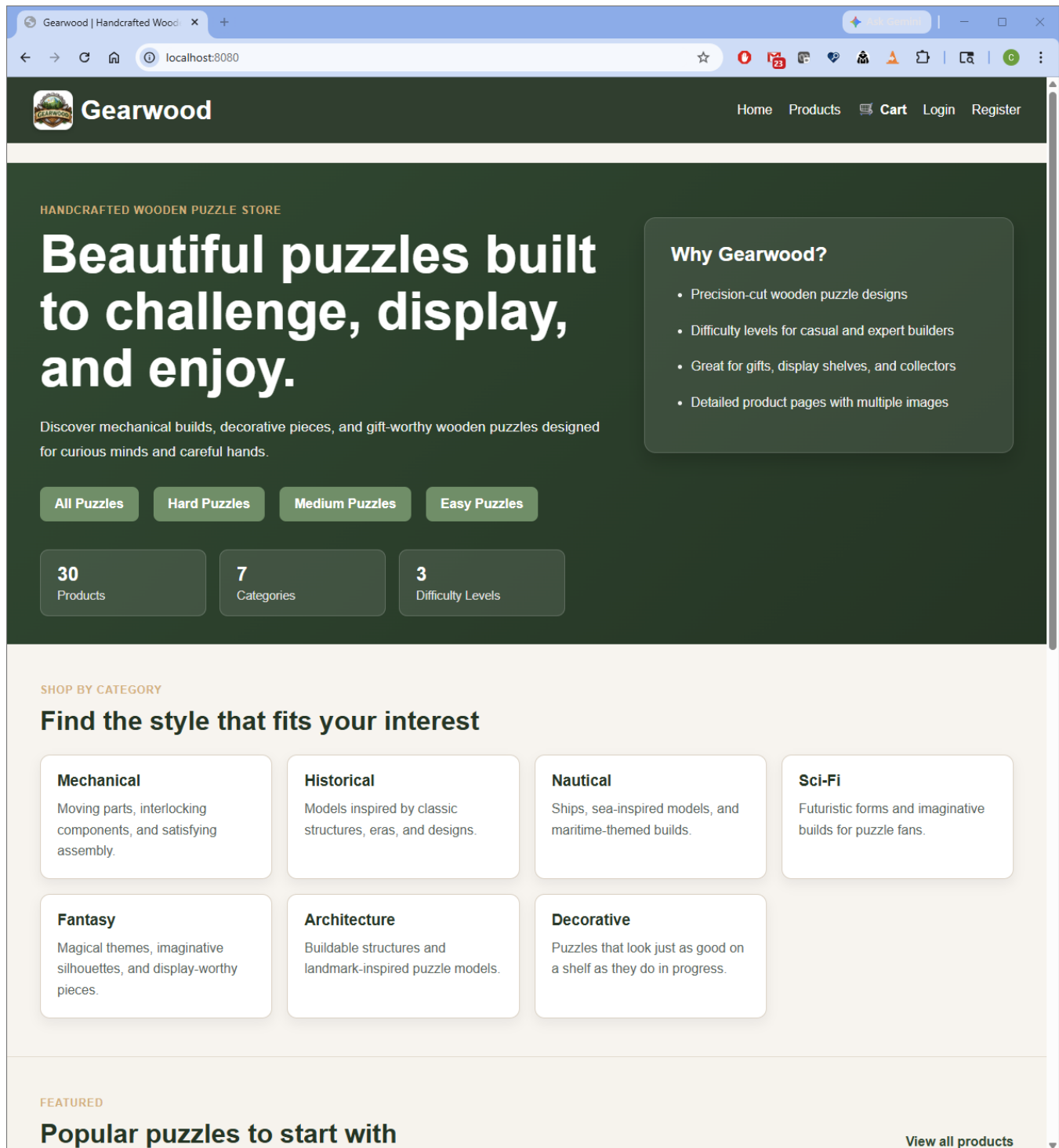
Suggested Screenshots

Students are encouraged to create a professional and visually organized user experience.

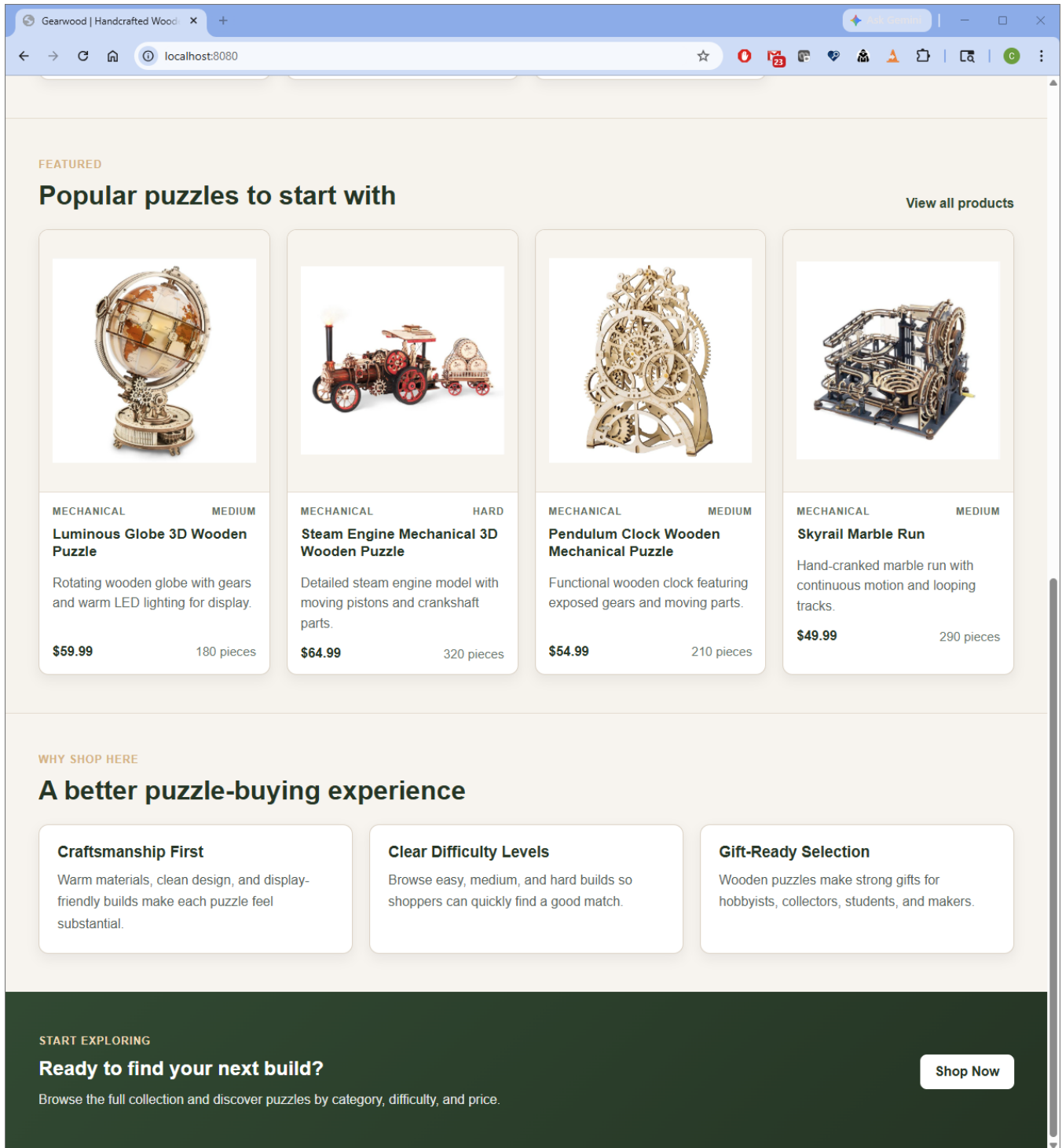
The exact visual appearance does not need to perfectly match the example screenshots.



A) Home page, user not logged in (upper-half)



B) Home page, user not logged in (lower-half)

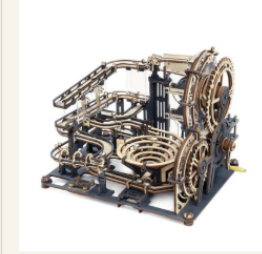


The screenshot shows a web browser window with the address bar displaying 'localhost:8080'. The website is titled 'Gearwood | Handcrafted Wood'.

FEATURED

Popular puzzles to start with

[View all products](#)

MECHANICAL	MEDIUM	MECHANICAL	HARD	MECHANICAL	MEDIUM	MECHANICAL	MEDIUM
							
Luminous Globe 3D Wooden Puzzle		Steam Engine Mechanical 3D Wooden Puzzle		Pendulum Clock Wooden Mechanical Puzzle		Skyrail Marble Run	
Rotating wooden globe with gears and warm LED lighting for display.		Detailed steam engine model with moving pistons and crankshaft parts.		Functional wooden clock featuring exposed gears and moving parts.		Hand-cranked marble run with continuous motion and looping tracks.	
\$59.99		\$64.99		\$54.99		\$49.99	
180 pieces		320 pieces		210 pieces		290 pieces	

WHY SHOP HERE

A better puzzle-buying experience

Craftsmanship First

Warm materials, clean design, and display-friendly builds make each puzzle feel substantial.

Clear Difficulty Levels

Browse easy, medium, and hard builds so shoppers can quickly find a good match.

Gift-Ready Selection

Wooden puzzles make strong gifts for hobbyists, collectors, students, and makers.

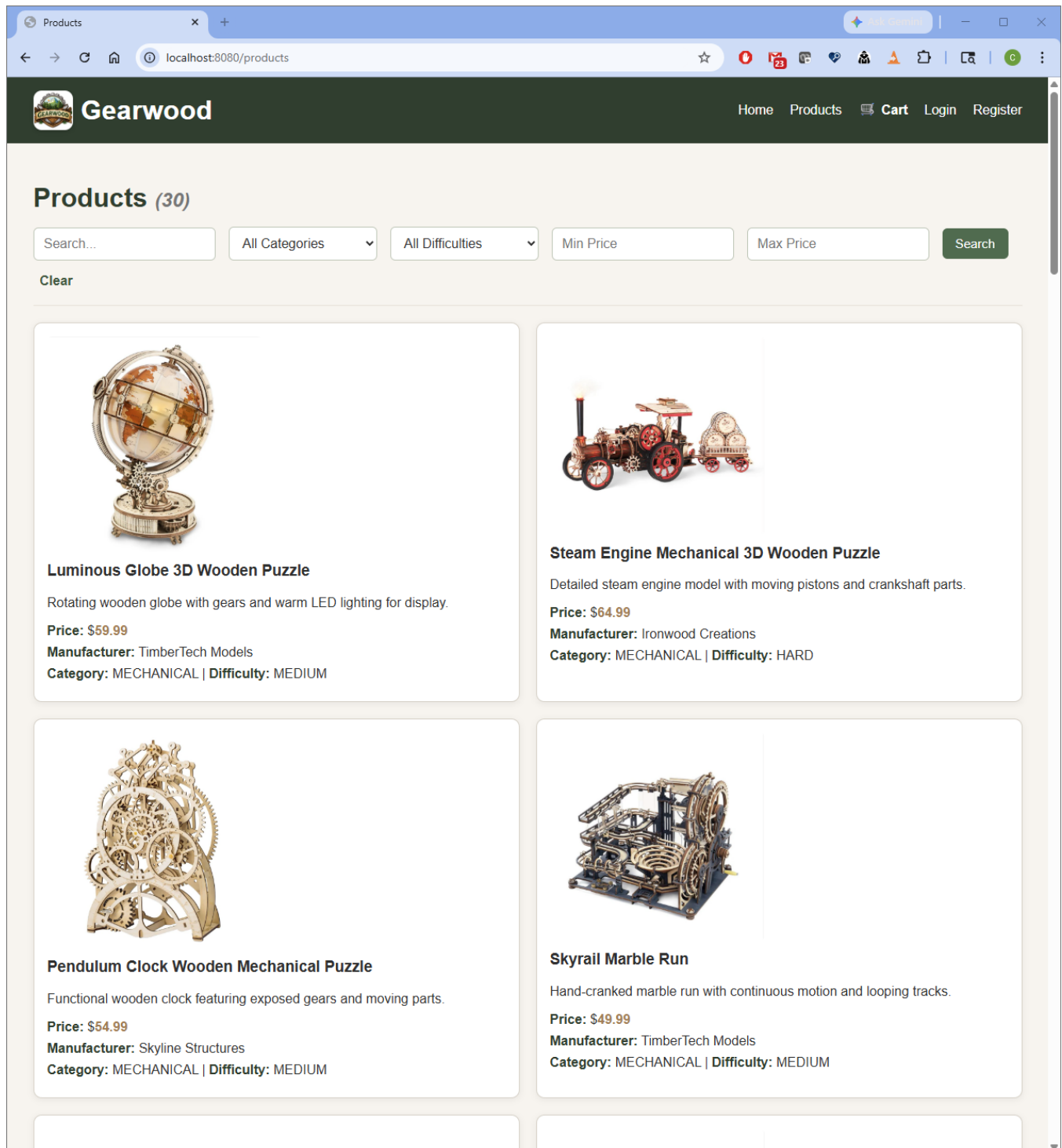
START EXPLORING

Ready to find your next build?

Browse the full collection and discover puzzles by category, difficulty, and price.

[Shop Now](#)

C) Browse Products (user not logged in)



The screenshot displays the 'Products' page of the Gearwood website. The page features a dark green header with the Gearwood logo and navigation links: Home, Products, Cart, Login, and Register. Below the header, the 'Products (30)' section is visible, including a search bar and filters for categories and difficulties. The main content area shows four product cards arranged in a 2x2 grid.

Product Name	Description	Price	Manufacturer	Category Difficulty
Luminous Globe 3D Wooden Puzzle	Rotating wooden globe with gears and warm LED lighting for display.	\$59.99	TimberTech Models	MECHANICAL MEDIUM
Steam Engine Mechanical 3D Wooden Puzzle	Detailed steam engine model with moving pistons and crankshaft parts.	\$64.99	Ironwood Creations	MECHANICAL HARD
Pendulum Clock Wooden Mechanical Puzzle	Functional wooden clock featuring exposed gears and moving parts.	\$54.99	Skyline Structures	MECHANICAL MEDIUM
Skyrail Marble Run	Hand-cranked marble run with continuous motion and looping tracks.	\$49.99	TimberTech Models	MECHANICAL MEDIUM

D) View Product Details (user not logged in) – primary image displayed

Product Details
localhost:8080/products/1


Gearwood
Home Products **Cart** Login Register

[← Back to Products](#)



Luminous Globe 3D Wooden Puzzle

Rotating wooden globe with gears and warm LED lighting for display.

Price: \$59.99

Manufacturer: TimberTech Models

Pieces: 180

Difficulty: MEDIUM

Category: MECHANICAL

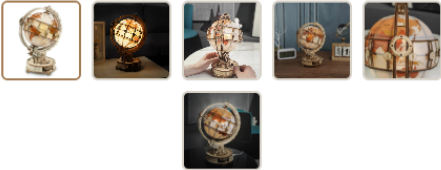
Add to Cart

Please log in to add this item to your cart. [Login to Add](#)

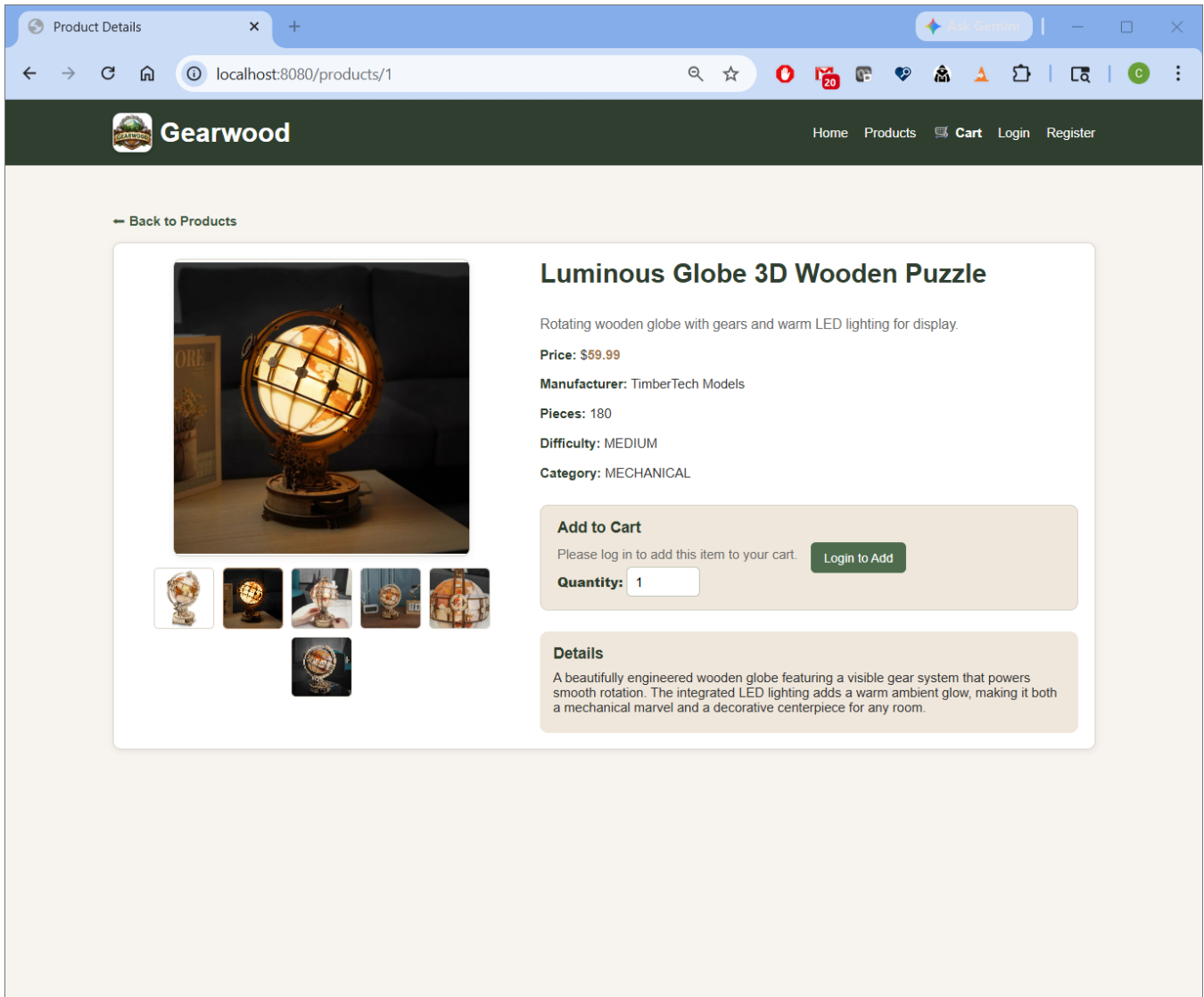
Quantity:

Details

A beautifully engineered wooden globe featuring a visible gear system that powers smooth rotation. The integrated LED lighting adds a warm ambient glow, making it both a mechanical marvel and a decorative centerpiece for any room.



E) View Product Details (user not logged in) – alternate (2nd) image displayed



Product Details

localhost:8080/products/1

Gearwood Home Products Cart Login Register

[← Back to Products](#)

Luminous Globe 3D Wooden Puzzle

Rotating wooden globe with gears and warm LED lighting for display.

Price: \$59.99

Manufacturer: TimberTech Models

Pieces: 180

Difficulty: MEDIUM

Category: MECHANICAL

Add to Cart

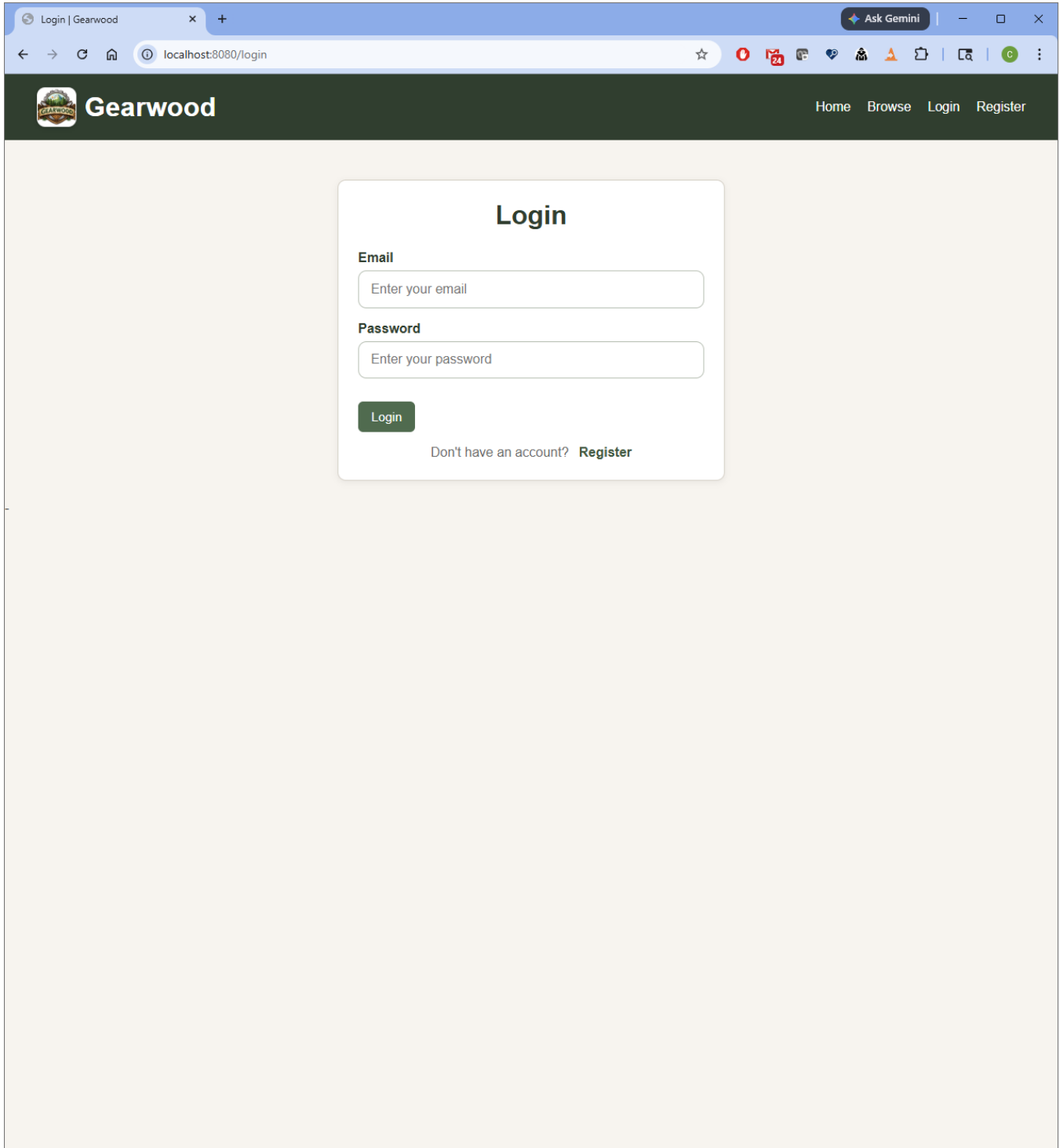
Please log in to add this item to your cart. [Login to Add](#)

Quantity: 1

Details

A beautifully engineered wooden globe featuring a visible gear system that powers smooth rotation. The integrated LED lighting adds a warm ambient glow, making it both a mechanical marvel and a decorative centerpiece for any room.

F) Login page



The screenshot shows a web browser window with the address bar displaying "localhost:8080/login". The page has a dark green header with the "Gearwood" logo and navigation links: "Home", "Browse", "Login", and "Register". The main content area is a light beige color. In the center, there is a white login form with a dark green "Login" button. Below the button, there is a link that says "Don't have an account? Register".

Login

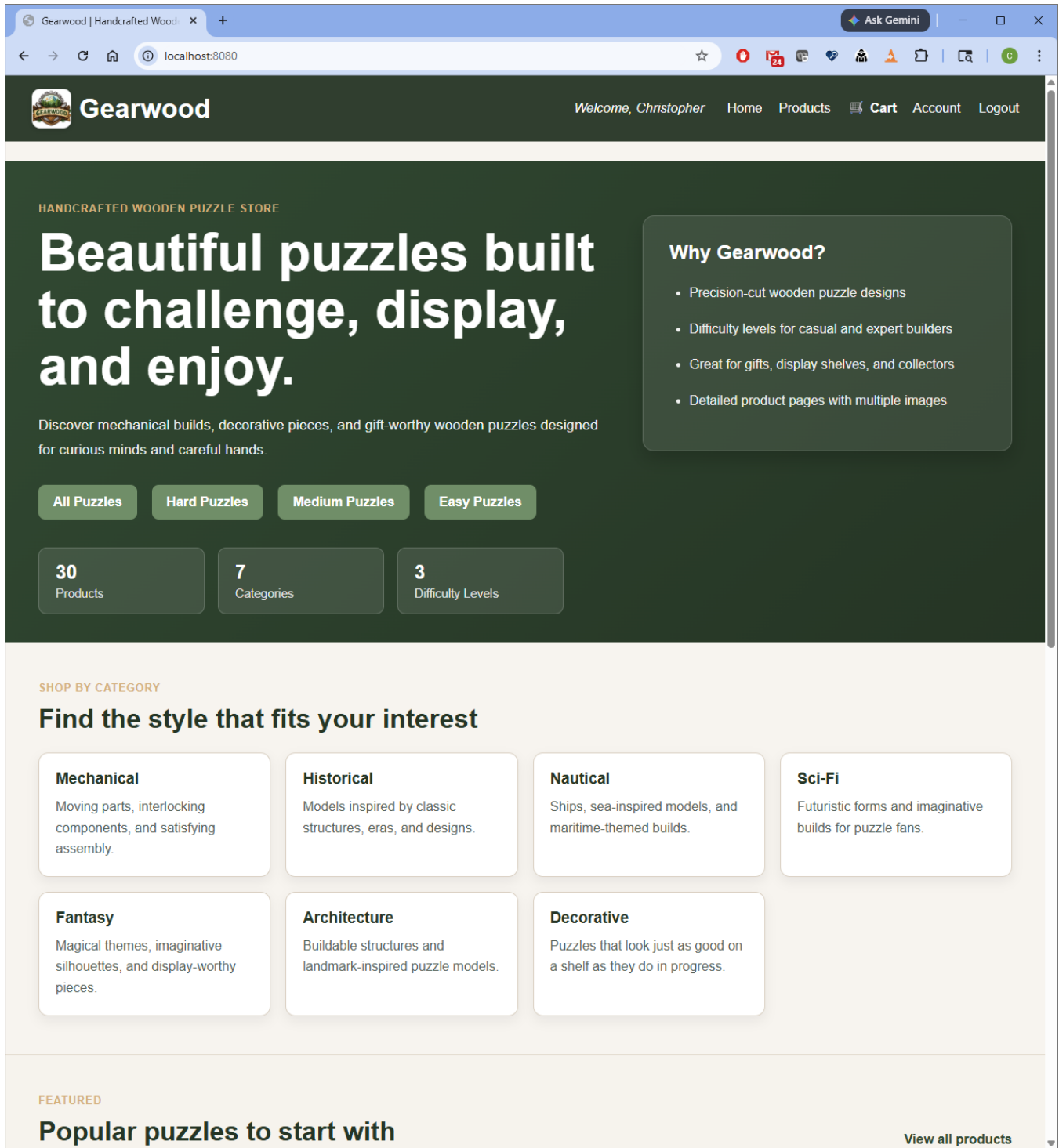
Email

Password

Login

Don't have an account? [Register](#)

G) Home page, user logged in



The screenshot shows the Gearwood website home page. The browser address bar indicates the URL is localhost:8080. The website header includes the Gearwood logo, a welcome message for Christopher, and navigation links for Home, Products, Cart, Account, and Logout. The main content area features a large hero section with the text "Beautiful puzzles built to challenge, display, and enjoy." and a sub-header "HANDCRAFTED WOODEN PUZZLE STORE". Below this, there are buttons for "All Puzzles", "Hard Puzzles", "Medium Puzzles", and "Easy Puzzles". A statistics section shows 30 Products, 7 Categories, and 3 Difficulty Levels. A "SHOP BY CATEGORY" section titled "Find the style that fits your interest" lists seven categories: Mechanical, Historical, Nautical, Sci-Fi, Fantasy, Architecture, and Decorative, each with a brief description. At the bottom, a "FEATURED" section titled "Popular puzzles to start with" includes a "View all products" link.

Gearwood | Handcrafted Wooden Puzzles

Welcome, Christopher | Home | Products | **Cart** | Account | Logout

HANDCRAFTED WOODEN PUZZLE STORE

Beautiful puzzles built to challenge, display, and enjoy.

Discover mechanical builds, decorative pieces, and gift-worthy wooden puzzles designed for curious minds and careful hands.

Why Gearwood?

- Precision-cut wooden puzzle designs
- Difficulty levels for casual and expert builders
- Great for gifts, display shelves, and collectors
- Detailed product pages with multiple images

All Puzzles | **Hard Puzzles** | **Medium Puzzles** | **Easy Puzzles**

30 Products | **7** Categories | **3** Difficulty Levels

SHOP BY CATEGORY

Find the style that fits your interest

Mechanical
Moving parts, interlocking components, and satisfying assembly.

Historical
Models inspired by classic structures, eras, and designs.

Nautical
Ships, sea-inspired models, and maritime-themed builds.

Sci-Fi
Futuristic forms and imaginative builds for puzzle fans.

Fantasy
Magical themes, imaginative silhouettes, and display-worthy pieces.

Architecture
Buildable structures and landmark-inspired puzzle models.

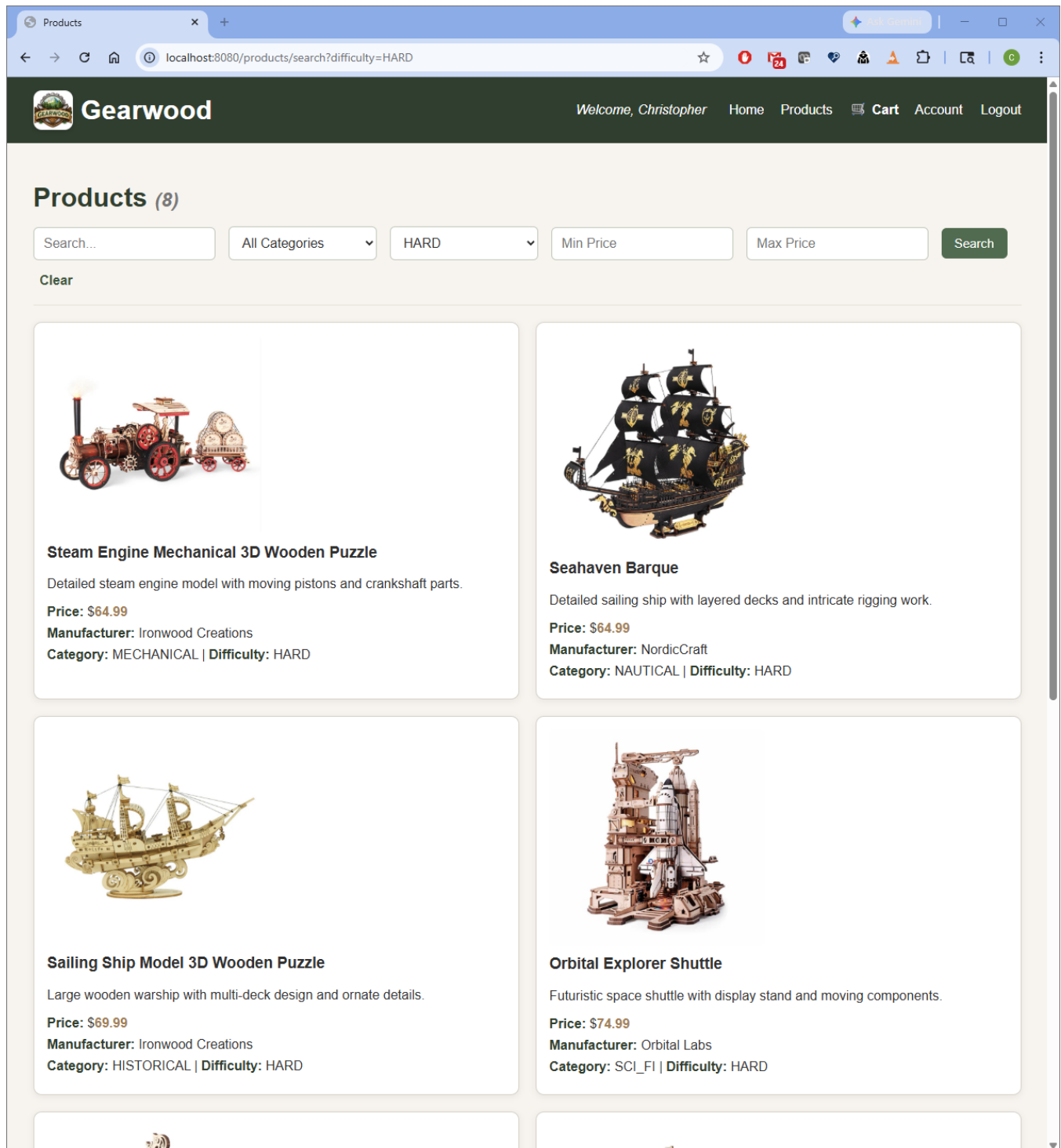
Decorative
Puzzles that look just as good on a shelf as they do in progress.

FEATURED

Popular puzzles to start with

[View all products](#)

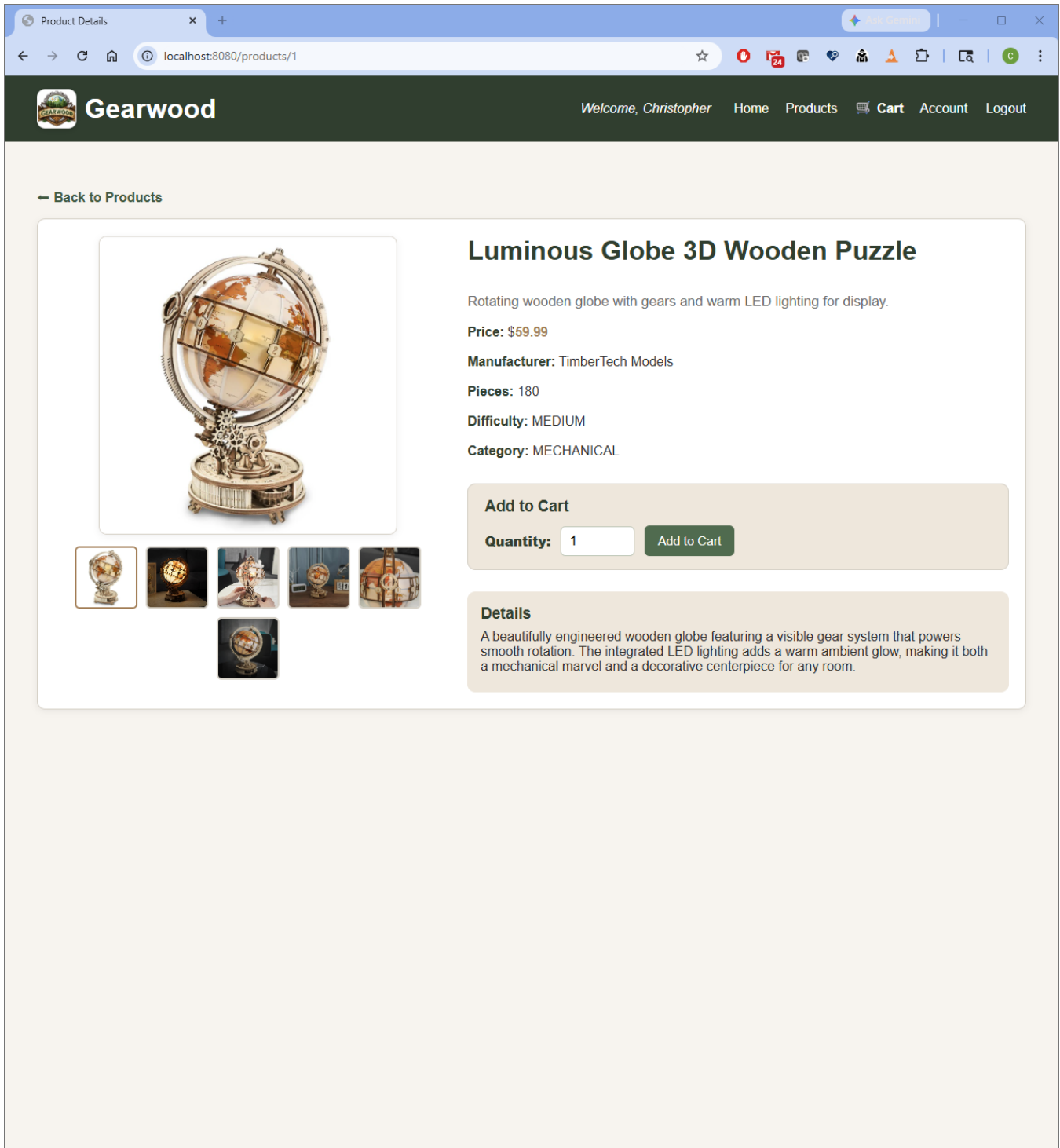
H) Browse Products (user logged in), search filters selected.



The screenshot shows a web browser window with the URL `localhost:8080/products/search?difficulty=HARD`. The website is 'Gearwood' and the user is logged in as 'Christopher'. The navigation bar includes links for Home, Products, Cart, Account, and Logout. The main content area is titled 'Products (8)' and features a search bar and filters. The search filters are set to 'All Categories' and 'HARD'. The search results are displayed in a grid of four product cards.

Product Name	Description	Price	Manufacturer	Category	Difficulty
Steam Engine Mechanical 3D Wooden Puzzle	Detailed steam engine model with moving pistons and crankshaft parts.	\$64.99	Ironwood Creations	MECHANICAL	HARD
Seahaven Barque	Detailed sailing ship with layered decks and intricate rigging work.	\$64.99	NordicCraft	NAUTICAL	HARD
Sailing Ship Model 3D Wooden Puzzle	Large wooden warship with multi-deck design and ornate details.	\$69.99	Ironwood Creations	HISTORICAL	HARD
Orbital Explorer Shuttle	Futuristic space shuttle with display stand and moving components.	\$74.99	Orbital Labs	SCI_FI	HARD

I) View Product Details (user logged in) – primary image displayed



The screenshot shows a web browser window with the URL `localhost:8080/products/1`. The website is for 'Gearwood' and the user is logged in as 'Christopher'. The navigation bar includes links for Home, Products, Cart, Account, and Logout. The main content area displays the 'Luminous Globe 3D Wooden Puzzle' with a large primary image and several smaller thumbnail images below it. The product description includes the price (\$59.99), manufacturer (TimberTech Models), number of pieces (180), difficulty (MEDIUM), and category (MECHANICAL). There is an 'Add to Cart' button and a quantity selector set to 1. A 'Details' section provides a brief description of the product.

Product Details:

- Product Name:** Luminous Globe 3D Wooden Puzzle
- Description:** Rotating wooden globe with gears and warm LED lighting for display.
- Price:** \$59.99
- Manufacturer:** TimberTech Models
- Pieces:** 180
- Difficulty:** MEDIUM
- Category:** MECHANICAL

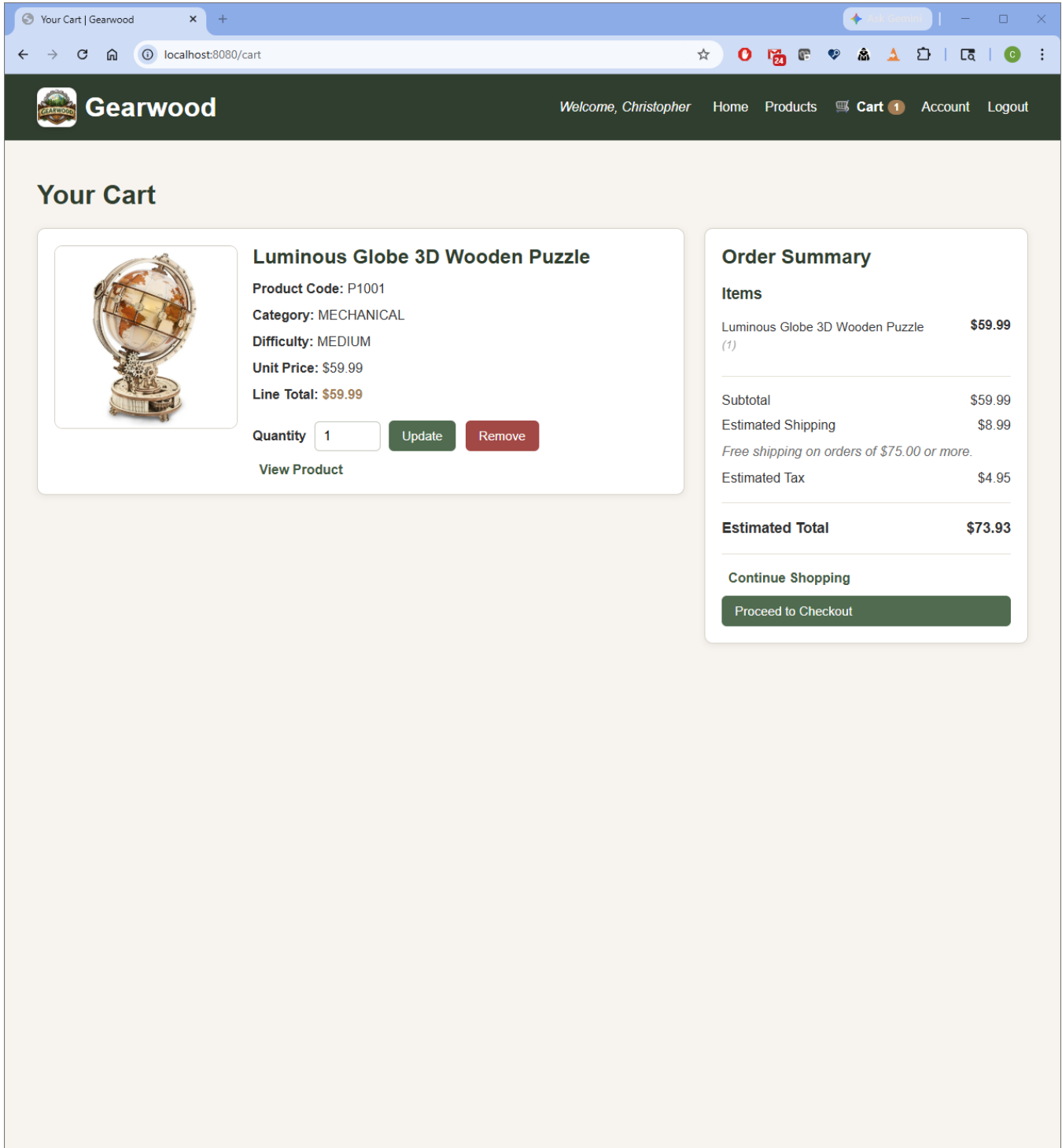
Add to Cart:

Quantity: Add to Cart

Details:

A beautifully engineered wooden globe featuring a visible gear system that powers smooth rotation. The integrated LED lighting adds a warm ambient glow, making it both a mechanical marvel and a decorative centerpiece for any room.

J) Cart Page (user logged in) – one item in the cart



Your Cart | Gearwood

Welcome, Christopher Home Products **Cart 1** Account Logout

Your Cart



Luminous Globe 3D Wooden Puzzle

Product Code: P1001

Category: MECHANICAL

Difficulty: MEDIUM

Unit Price: \$59.99

Line Total: **\$59.99**

Quantity Update Remove

[View Product](#)

Order Summary

Items

Luminous Globe 3D Wooden Puzzle (1)	\$59.99
-------------------------------------	----------------

Subtotal	\$59.99
Estimated Shipping	\$8.99
<i>Free shipping on orders of \$75.00 or more.</i>	
Estimated Tax	\$4.95

Estimated Total **\$73.93**

Continue Shopping

[Proceed to Checkout](#)

K) Cart Page (user logged in) – two items in the cart


Gearwood

Welcome, Christopher
Home
Products
Cart 2
Account
Logout

Your Cart



Luminous Globe 3D Wooden Puzzle
Product Code: P1001
Category: MECHANICAL
Difficulty: MEDIUM
Unit Price: \$59.99
Line Total: **\$59.99**

Quantity

[View Product](#)



Nordic Dragon Longship
Product Code: P1006
Category: HISTORICAL
Difficulty: MEDIUM
Unit Price: \$59.99
Line Total: **\$59.99**

Quantity

[View Product](#)

Order Summary

Items

Luminous Globe 3D Wooden Puzzle (1)	\$59.99
Nordic Dragon Longship (1)	\$59.99
Subtotal	\$119.98
Estimated Shipping	\$0.00
<i>You qualified for free shipping.</i>	
Estimated Tax	\$9.90
Estimated Total	\$129.88
Continue Shopping	
Proceed to Checkout	

L) Checkout Page (user logged in)

Checkout | Gearwood

localhost:8080/checkout


Gearwood

Welcome, Christopher
Home
Products
Cart 2
Account
Logout

[← Back to Cart](#)

Checkout

Shipping Information

Full Name

Address Line 1

Address Line 2

City

State

ZIP

Payment Information

Cardholder Name

Card Number

Expiration Month (MM)

Expiration Year (YYYY)

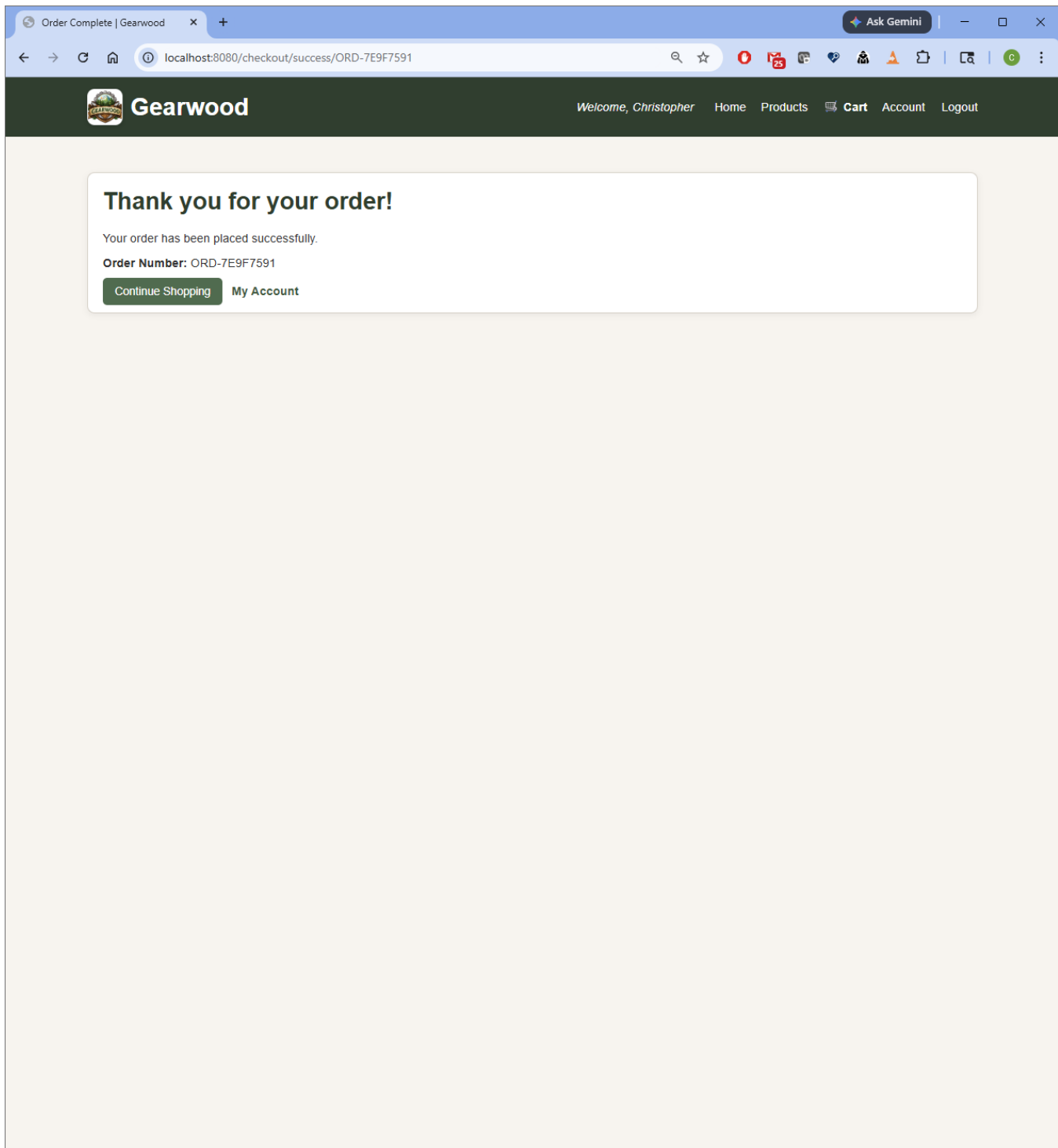
CVV

[Place Order](#)

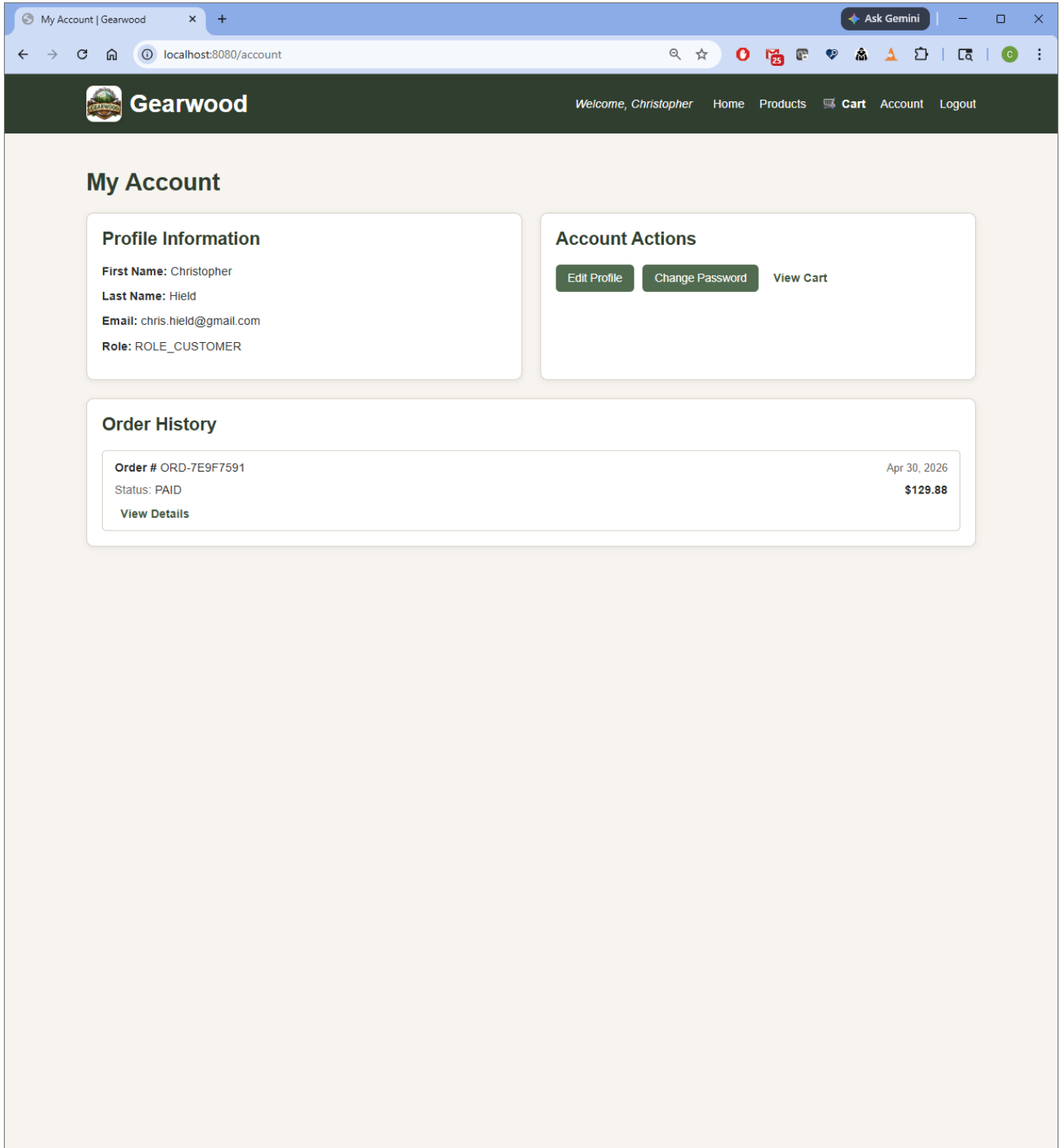
Order Summary

Luminous Globe 3D Wooden Puzzle <i>x1</i>	\$59.99
Nordic Dragon Longship <i>x1</i>	\$59.99
Subtotal	\$119.98
Shipping	\$0.00
<i>You qualified for free shipping.</i>	
Tax	\$9.90
Total	\$129.88

M) Checkout Success Page (user logged in)



N) Account Page (user logged in)



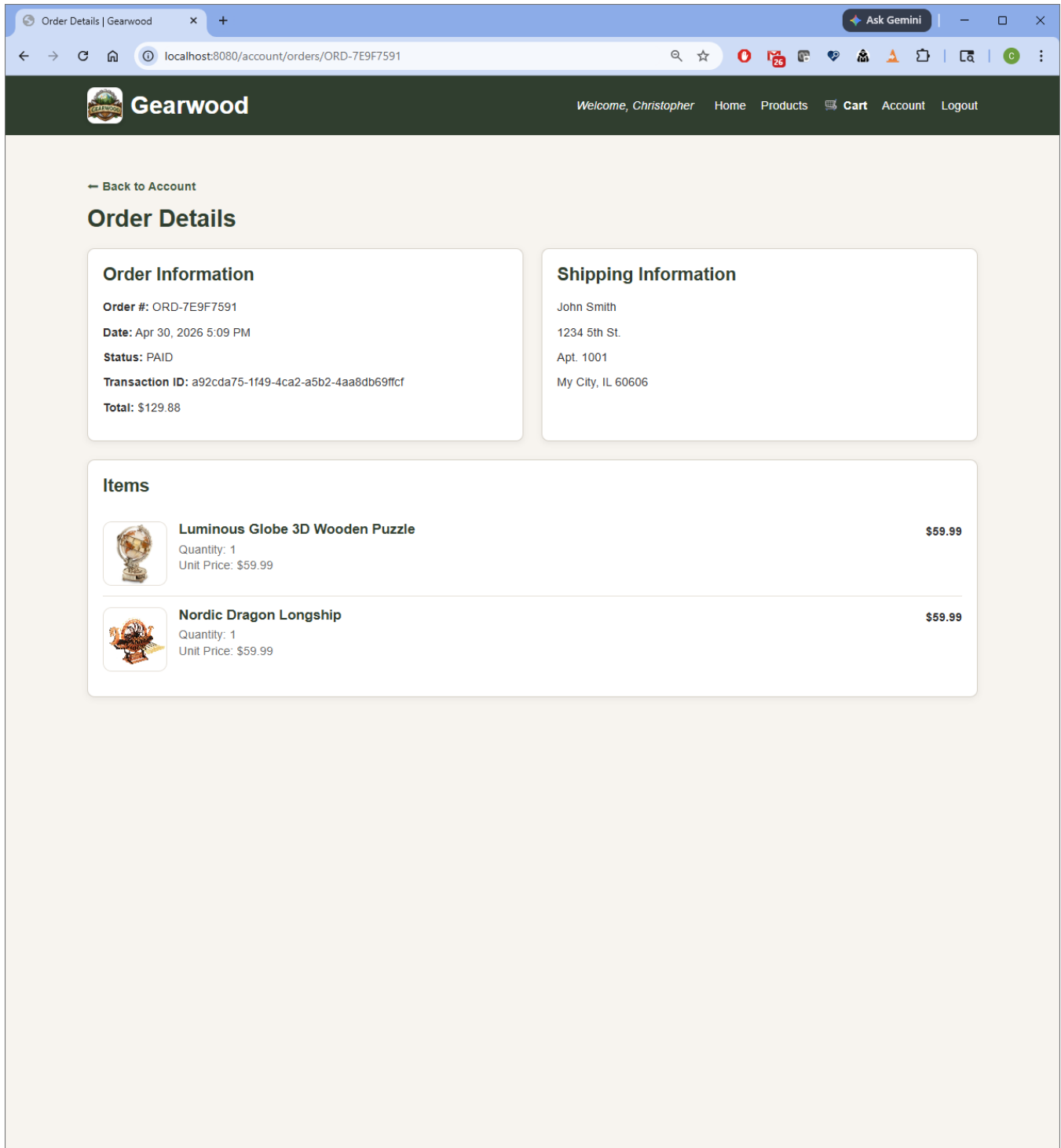
The screenshot shows a web browser window with the address bar displaying "localhost:8080/account". The browser's address bar also shows "My Account | Gearwood" and a search bar with "Ask Gemini". The browser's toolbar includes navigation buttons (back, forward, refresh, home), a search bar, and various extension icons (notifications, calendar, weather, etc.).

The page header is dark green with the Gearwood logo on the left and navigation links on the right: "Welcome, Christopher", "Home", "Products", "Cart", "Account", and "Logout".

The main content area is titled "My Account" and contains three sections:

- Profile Information**:
 - First Name:** Christopher
 - Last Name:** Hield
 - Email:** chris.hield@gmail.com
 - Role:** ROLE_CUSTOMER
- Account Actions**:
 - [Edit Profile](#)
 - [Change Password](#)
 - [View Cart](#)
- Order History**:
 - Order #** ORD-7E9F7591 Apr 30, 2026
 - Status:** PAID **\$129.88**
 - [View Details](#)

O) Order History Page (user logged in)



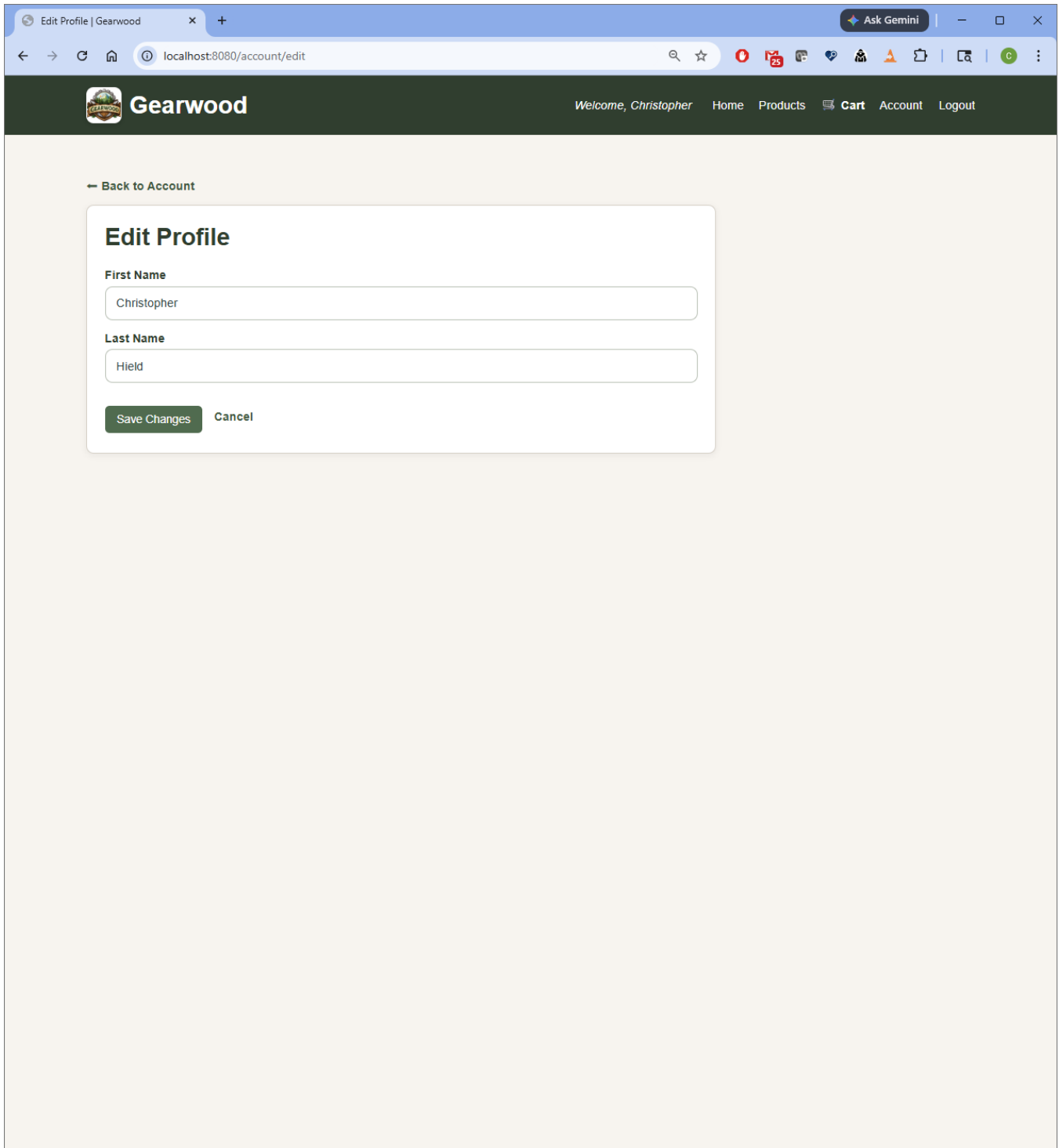
The screenshot shows a web browser window with the URL `localhost:8080/account/orders/ORD-7E9F7591`. The page is titled "Order Details" and features a navigation bar with the Gearwood logo and user information "Welcome, Christopher". The main content area includes a "Back to Account" link and two columns of information: "Order Information" and "Shipping Information". Below these is a table of items.

Order Information	
Order #:	ORD-7E9F7591
Date:	Apr 30, 2026 5:09 PM
Status:	PAID
Transaction ID:	a92cda75-1f49-4ca2-a5b2-4aa8db69ffcf
Total:	\$129.88

Shipping Information	
John Smith	
1234 5th St.	
Apt. 1001	
My City, IL 60606	

Items		
	Luminous Globe 3D Wooden Puzzle Quantity: 1 Unit Price: \$59.99	\$59.99
	Nordic Dragon Longship Quantity: 1 Unit Price: \$59.99	\$59.99

P) Edit Profile Page (user logged in)

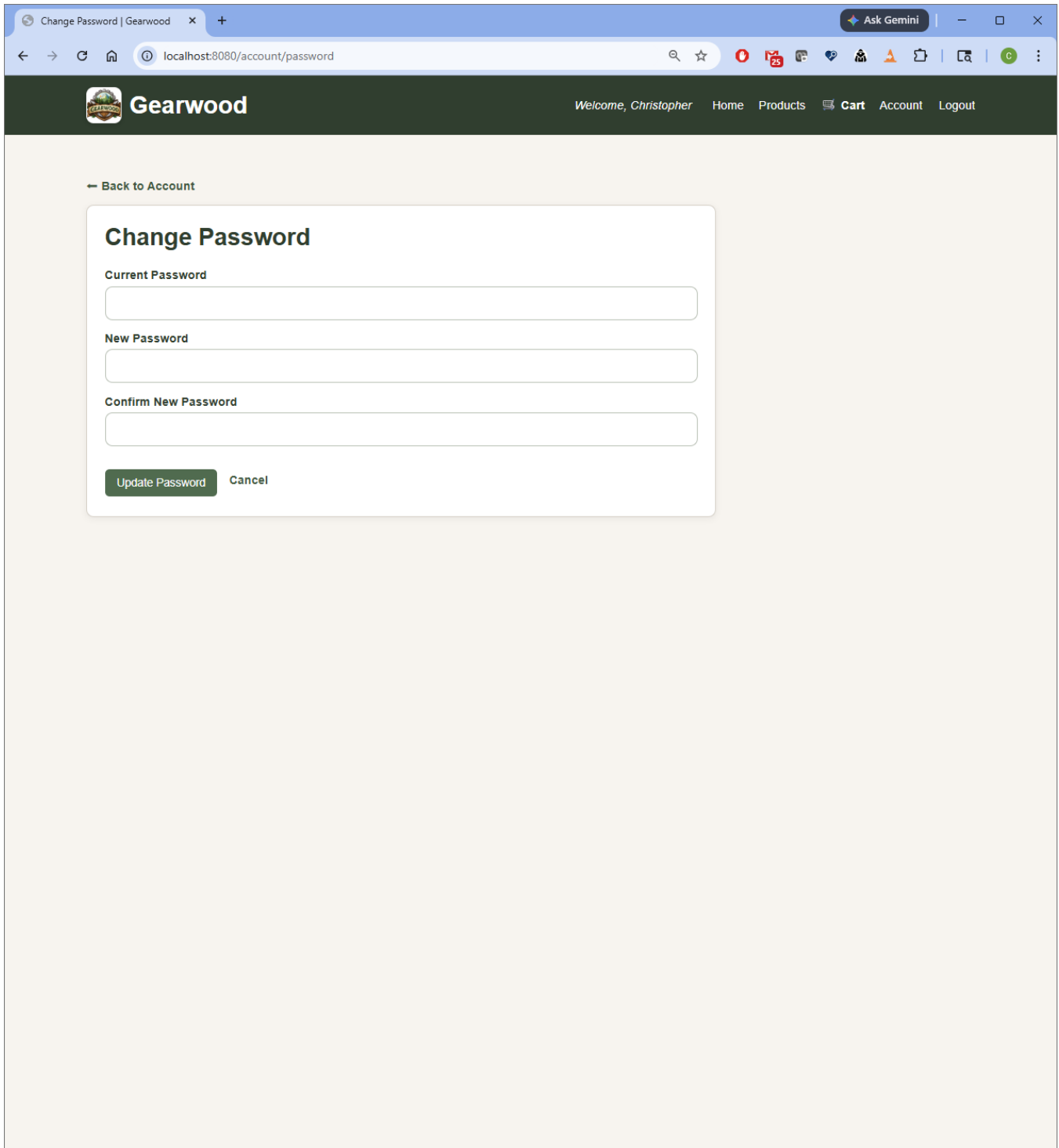


The screenshot shows a web browser window with the address bar displaying "localhost:8080/account/edit". The browser's address bar also shows "Edit Profile | Gearwood" and a search icon. The browser's toolbar includes icons for back, forward, home, and search, as well as a "Ask Gemini" button. The browser's status bar shows various system icons, including a clock, a calendar, and a battery level indicator.

The Gearwood website header is dark green and features the Gearwood logo on the left, the text "Welcome, Christopher" in the center, and navigation links for "Home", "Products", "Cart", "Account", and "Logout" on the right.

The main content area is light beige and contains a "Back to Account" link. Below this link is a white form titled "Edit Profile". The form contains two text input fields: "First Name" with the value "Christopher" and "Last Name" with the value "Hield". At the bottom of the form are two buttons: "Save Changes" (green) and "Cancel" (grey).

Q) Change Password Page (user logged in)



Change Password | Gearwood

localhost:8080/account/password

Gearwood Welcome, Christopher Home Products **Cart** Account Logout

[← Back to Account](#)

Change Password

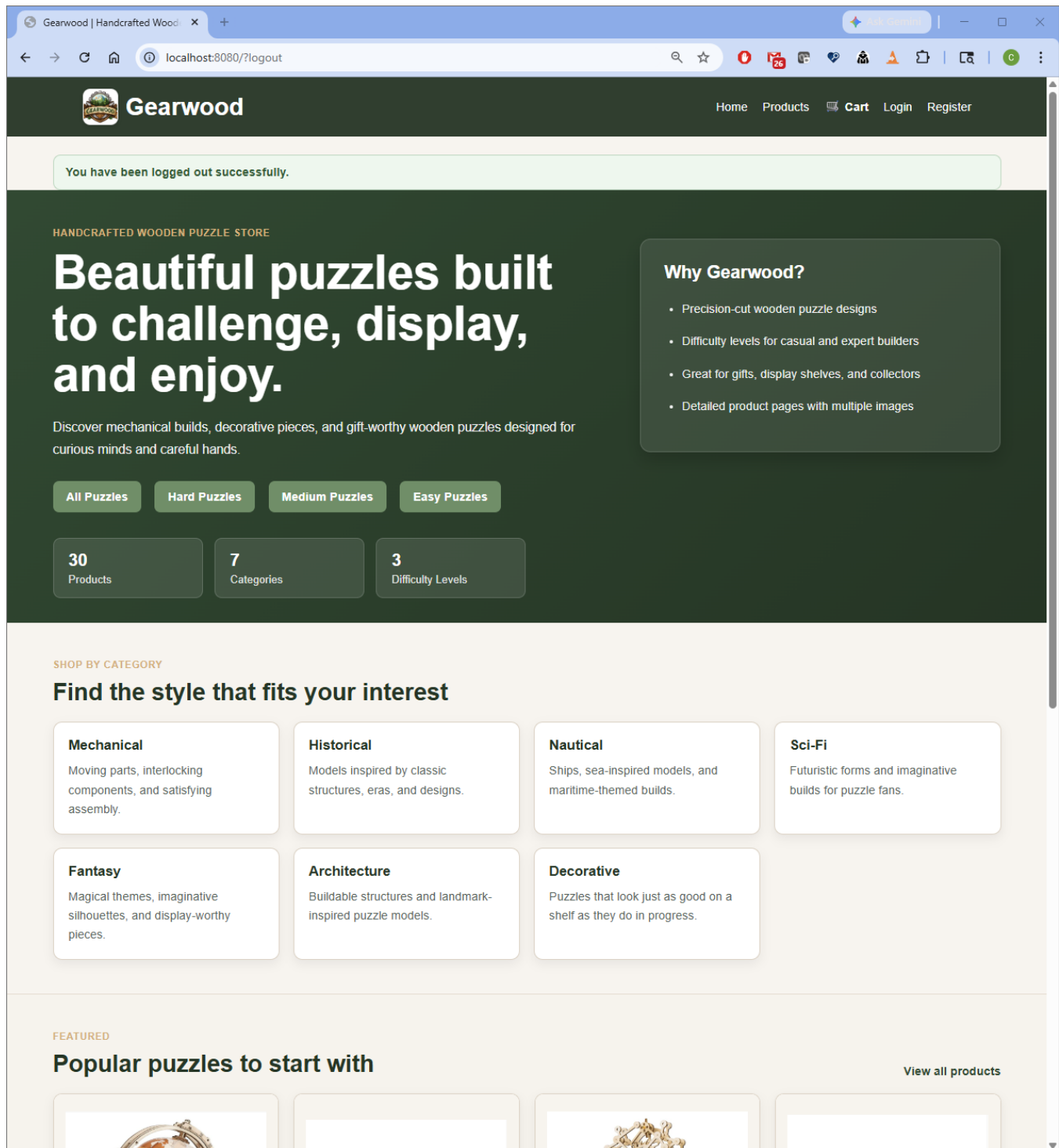
Current Password

New Password

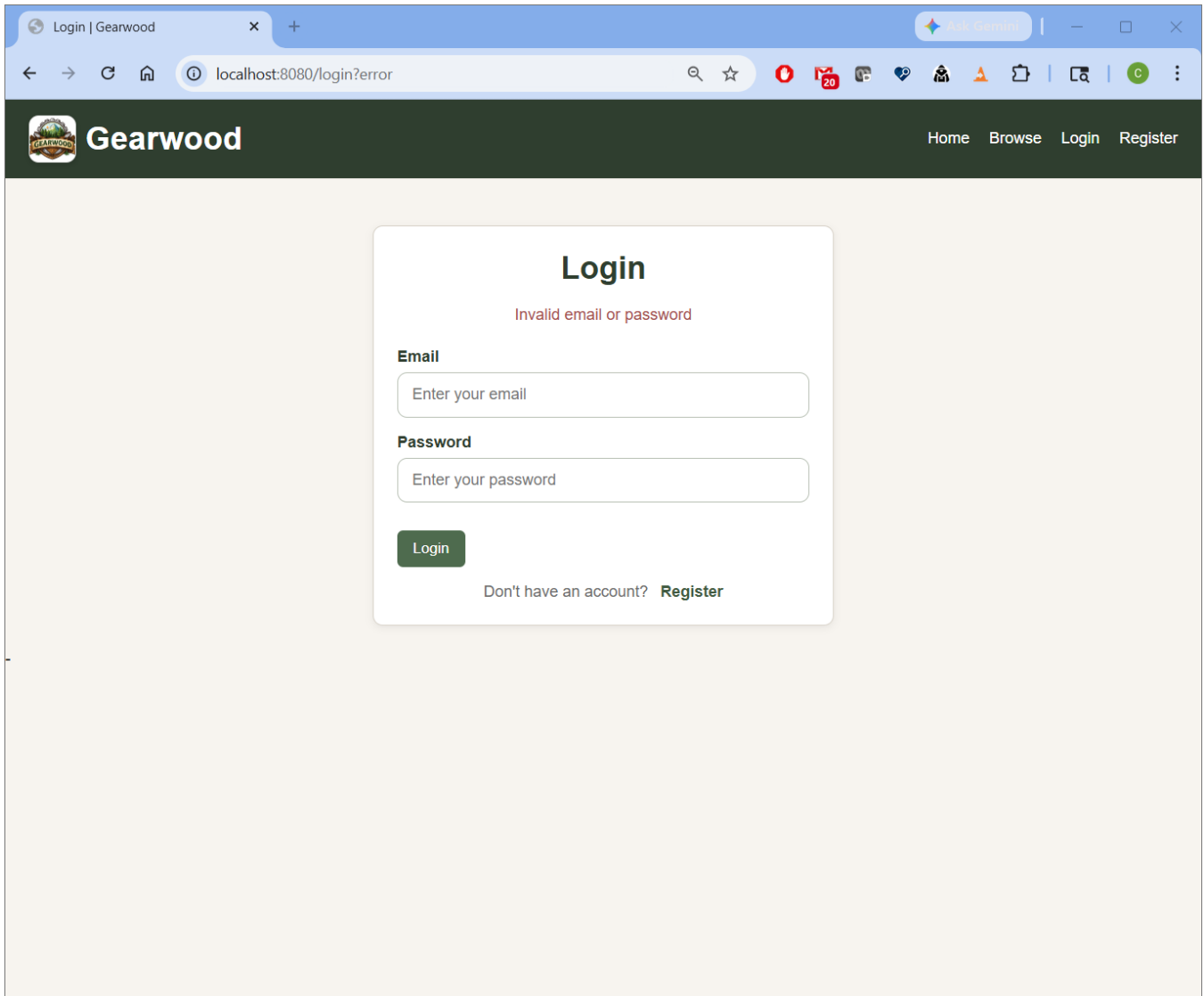
Confirm New Password

[Update Password](#) [Cancel](#)

R) Home page, showing successful logout message



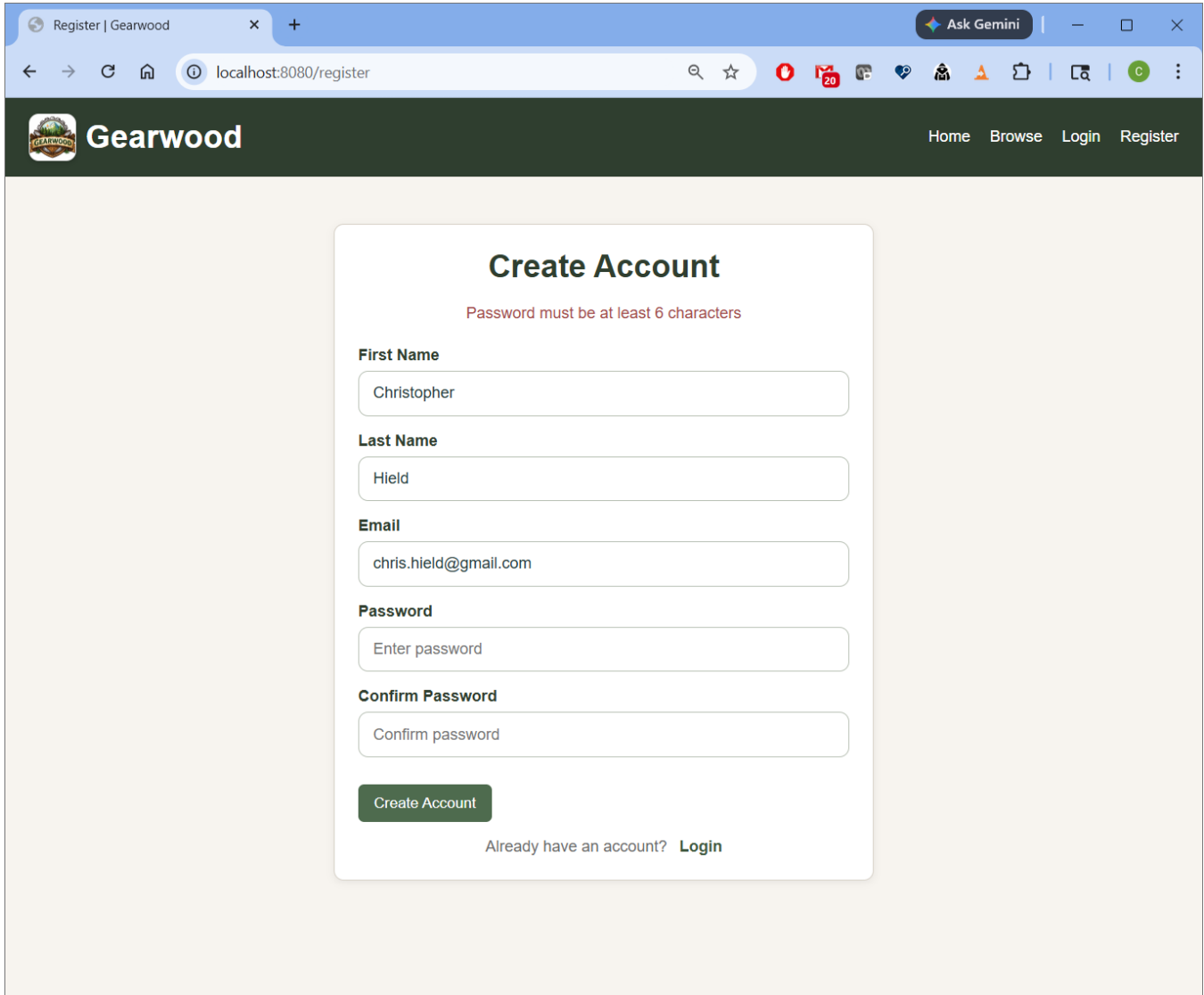
S) Unsuccessful login error message



The screenshot shows a web browser window with the address bar displaying "localhost:8080/login?error". The page title is "Login | Gearwood". The website header includes the "Gearwood" logo and navigation links: "Home", "Browse", "Login", and "Register". The main content area features a "Login" form with the following elements:

- Login** (Title)
- Invalid email or password** (Error message)
- Email** (Label)
- (Email input field)
- Password** (Label)
- (Password input field)
- (Login button)
- Don't have an account? [Register](#) (Link to register)

T) Unsuccessful account creation error message (one of many possible errors)



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/register'. The page title is 'Register | Gearwood'. The browser's address bar shows 'localhost:8080/register'. The page features a dark green header with the 'Gearwood' logo and navigation links: 'Home', 'Browse', 'Login', and 'Register'. The main content area is a light beige background with a white 'Create Account' form. The form has the title 'Create Account' and a red error message: 'Password must be at least 6 characters'. The form fields are: 'First Name' (Christopher), 'Last Name' (Hield), 'Email' (chris.hield@gmail.com), 'Password' (Enter password), and 'Confirm Password' (Confirm password). A green 'Create Account' button is at the bottom of the form. Below the button, it says 'Already have an account? Login'.

Register | Gearwood

Ask Gemini

localhost:8080/register

Gearwood Home Browse Login Register

Create Account

Password must be at least 6 characters

First Name

Christopher

Last Name

Hield

Email

chris.hield@gmail.com

Password

Enter password

Confirm Password

Confirm password

Create Account

Already have an account? [Login](#)

U) Unsuccessful checkout payment error message

Checkout | Gearwood

localhost:8080/checkout


Gearwood

Welcome, Christopher
Home
Products
Cart 2
Account
Logout

[← Back to Cart](#)

Checkout

The payment was declined by the payment processor.

Shipping Information

Full Name

Address Line 1

Address Line 2

City

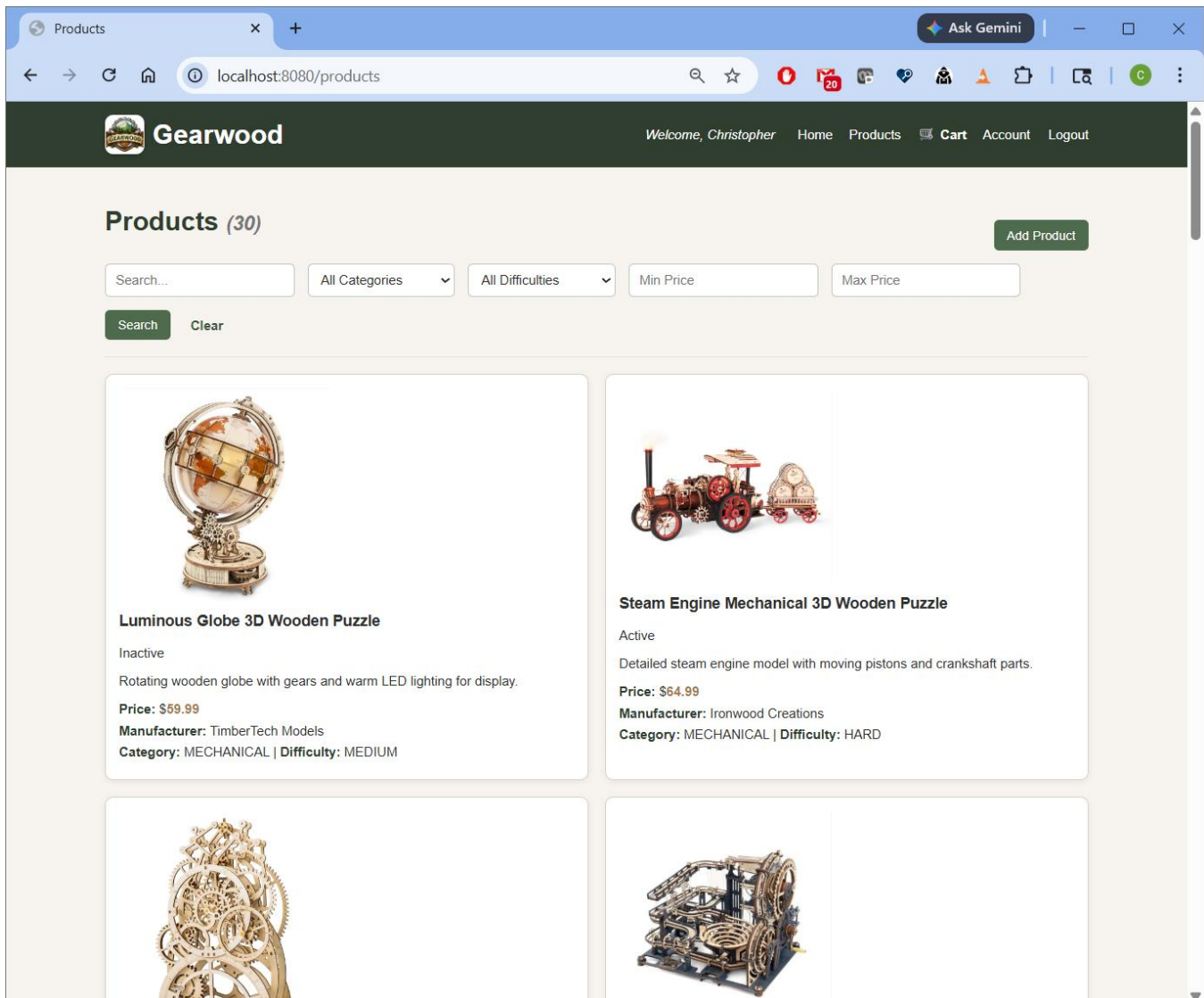
State

ZIP

Order Summary

Velocity Racer Car x1	\$39.99
Sailing Ship Model 3D Wooden Puzzle x1	\$69.99
Subtotal	\$109.98
Shipping	\$0.00
<i>You qualified for free shipping.</i>	
Tax	\$9.07
Total	\$119.05

V) Admin user logged in; Active/Inactive status visible in browse products page



The screenshot shows a web browser window with the URL `localhost:8080/products`. The website is 'Gearwood' and the user is logged in as 'Christopher'. The 'Products' page displays a grid of 30 products. The first two products are visible:

- Luminous Globe 3D Wooden Puzzle**: Inactive. Rotating wooden globe with gears and warm LED lighting for display. Price: \$59.99. Manufacturer: TimberTech Models. Category: MECHANICAL | Difficulty: MEDIUM.
- Steam Engine Mechanical 3D Wooden Puzzle**: Active. Detailed steam engine model with moving pistons and crankshaft parts. Price: \$64.99. Manufacturer: Ironwood Creations. Category: MECHANICAL | Difficulty: HARD.

W) Admin user logged in; Product details page, Edit Product & Deactivate Product buttons visible

Product Details

Ask Gemini

localhost:8080/products/1


Gearwood

Welcome, Christopher

Home

Products

Cart

Account

Logout

← Back to Products



Luminous Globe 3D Wooden Puzzle

Rotating wooden globe with gears and warm LED lighting for display.

Price: \$59.99

Manufacturer: TimberTech Models

Pieces: 180

Difficulty: MEDIUM

Category: MECHANICAL

Edit Product
Deactivate Product







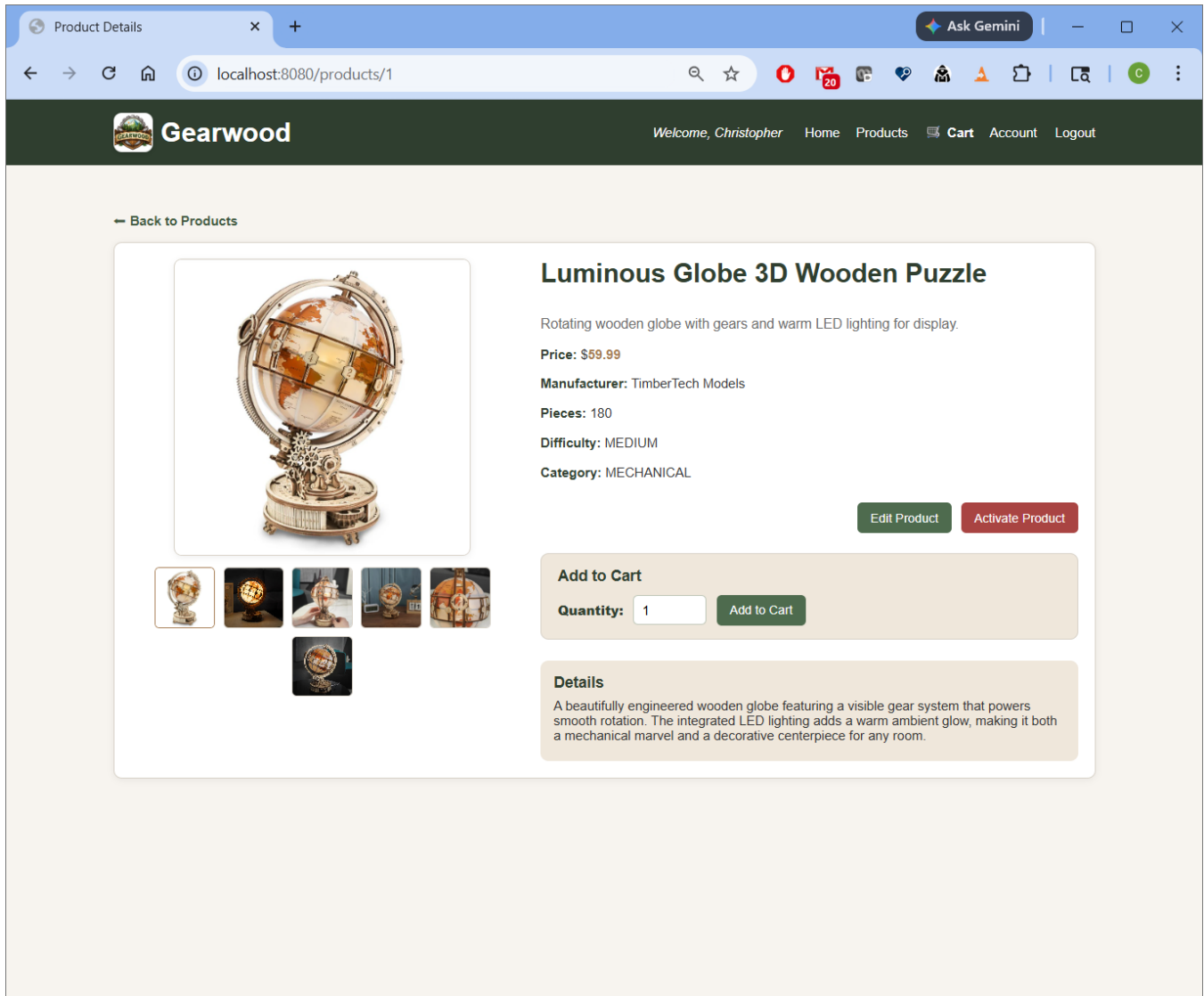

Add to Cart

Quantity: Add to Cart

Details

A beautifully engineered wooden globe featuring a visible gear system that powers smooth rotation. The integrated LED lighting adds a warm ambient glow, making it both a mechanical marvel and a decorative centerpiece for any room.

X) Admin user logged in; Product details page, Edit Product & Activate Product buttons visible

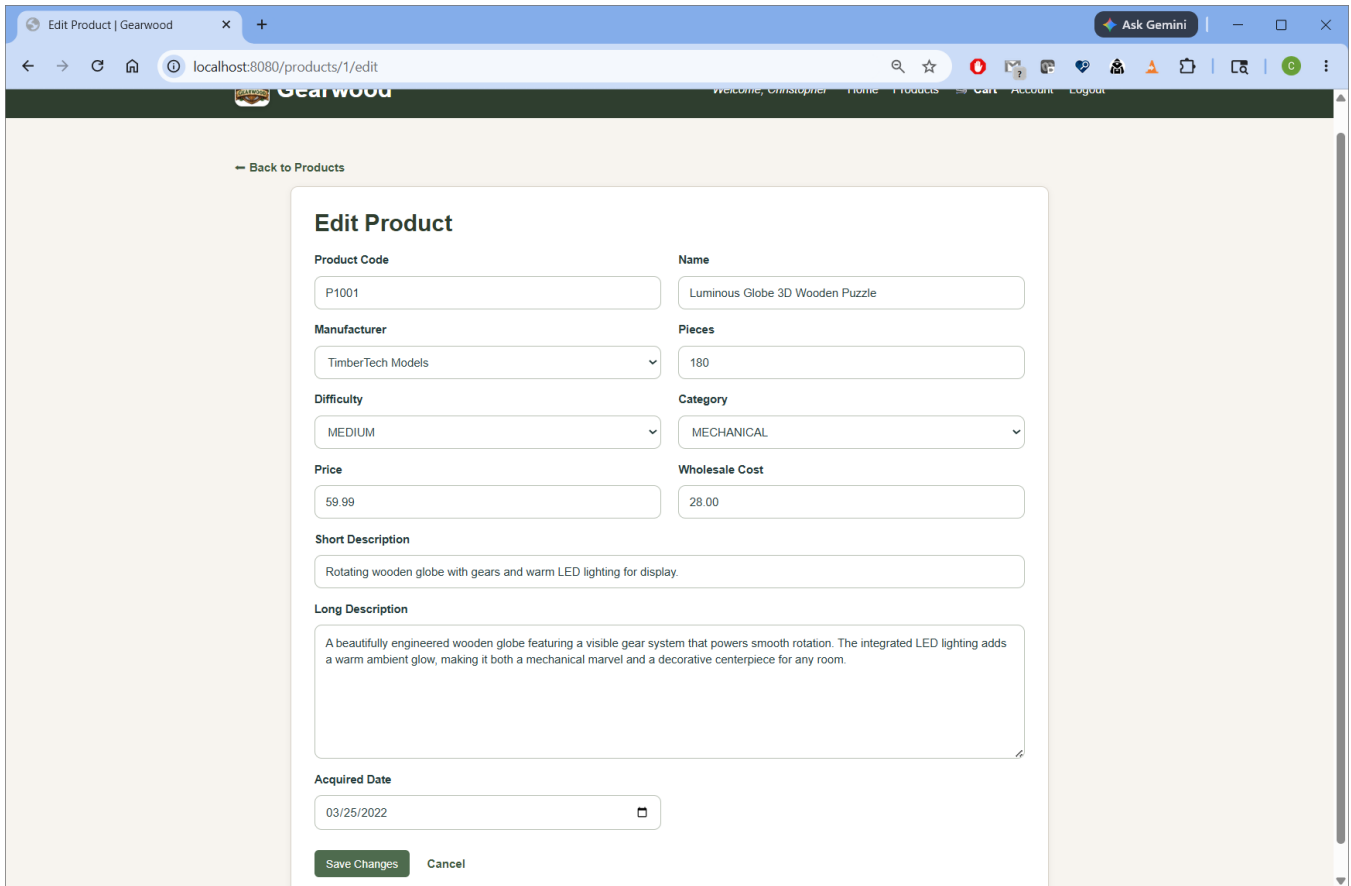


The screenshot shows a web browser window with the URL `localhost:8080/products/1`. The page is titled "Product Details" and features the Gearwood logo and navigation links: "Welcome, Christopher", "Home", "Products", "Cart", "Account", and "Logout".

The main content area displays the "Luminous Globe 3D Wooden Puzzle". The product description states: "Rotating wooden globe with gears and warm LED lighting for display." The price is listed as \$59.99, the manufacturer is TimberTech Models, there are 180 pieces, the difficulty is MEDIUM, and the category is MECHANICAL.

Below the product information, there are two buttons: "Edit Product" (green) and "Activate Product" (red). Below these buttons is an "Add to Cart" section with a quantity input field set to 1 and an "Add to Cart" button (green). A "Details" section follows, providing a description of the product: "A beautifully engineered wooden globe featuring a visible gear system that powers smooth rotation. The integrated LED lighting adds a warm ambient glow, making it both a mechanical marvel and a decorative centerpiece for any room."

Y) Admin user logged in; Edit product page



Browser tabs: Edit Product | Gearwood

Address bar: localhost:8080/products/1/edit

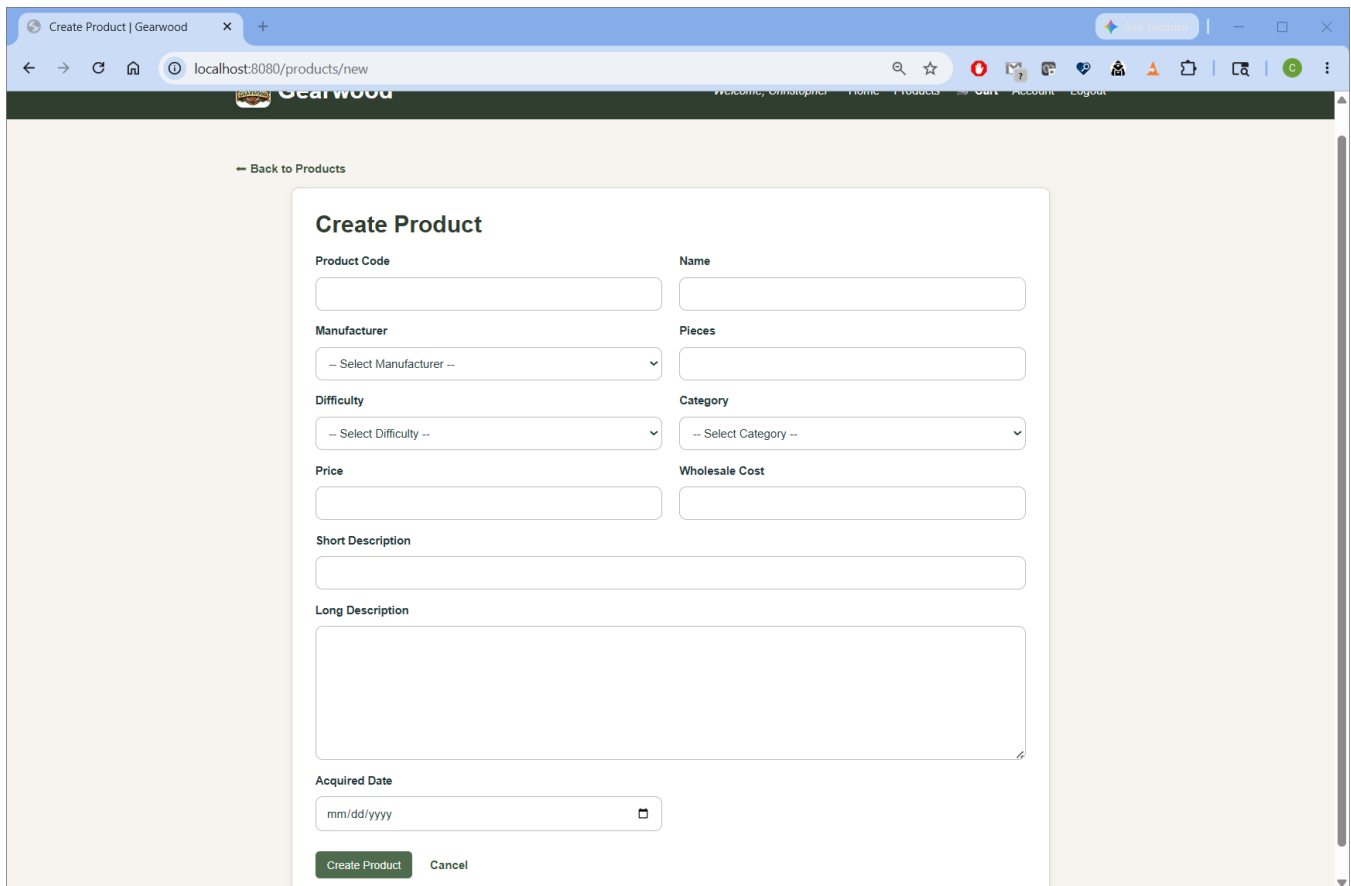
Navigation: Welcome, Christopher | Home | Products | Cart | Account | Logout

Back to Products

Edit Product

Product Code	Name
P1001	Luminous Globe 3D Wooden Puzzle
Manufacturer	Pieces
TimberTech Models	180
Difficulty	Category
MEDIUM	MECHANICAL
Price	Wholesale Cost
59.99	28.00
Short Description	
Rotating wooden globe with gears and warm LED lighting for display.	
Long Description	
A beautifully engineered wooden globe featuring a visible gear system that powers smooth rotation. The integrated LED lighting adds a warm ambient glow, making it both a mechanical marvel and a decorative centerpiece for any room.	
Acquired Date	
03/25/2022	
Save Changes Cancel	

Z) Admin user logged in; Create product page



← Back to Products

Create Product

Product Code	Name
Manufacturer	Pieces
-- Select Manufacturer --	
Difficulty	Category
-- Select Difficulty --	-- Select Category --
Price	Wholesale Cost
Short Description	
Long Description	
Acquired Date	
mm/dd/yyyy	